

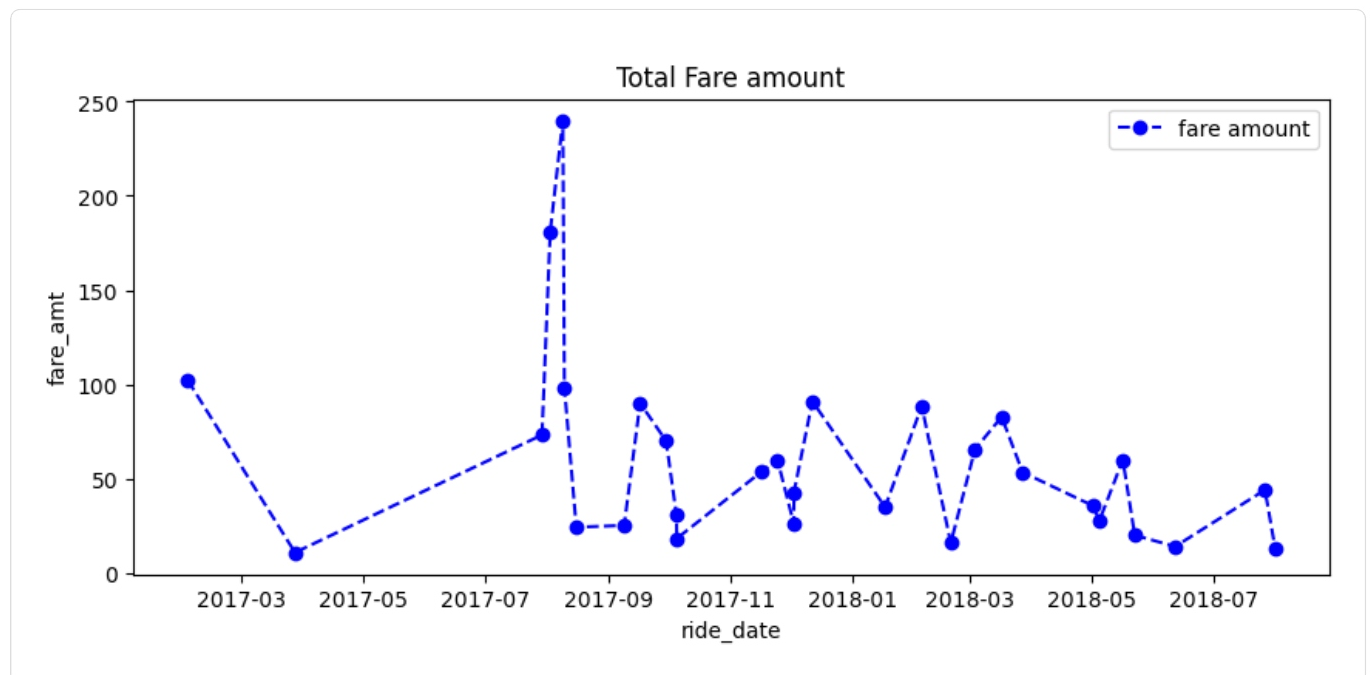
Visualization Assessment Solution Report

Question 1: 1. Line Chart (Trend)

Your Code:

```
import matplotlib.pyplot as plt
fare_amt = merged_df.groupby('ride_date').agg({'fare_amt':sum}).reset_index()
# Write your plotting code here:
plt.figure(figsize=(10,4))
# ...
# Group by purchase date (as string or period) to get daily order counts
plt.plot(fare_amt['ride_date'], fare_amt['fare_amt'],
color='blue', linestyle='--', marker='o', label='fare amount')
plt.title("Total Fare amount")
plt.xlabel("ride_date")
plt.ylabel("fare_amt")
plt.legend()
plt.show()
```

Rendered Output:



Question 2: 2. Bar Chart (Categorical)

Your Code:

```
# Write your plotting code here:
import seaborn as sns
sns.barplot(x=merged_df.city, y=merged_df.fare_amt, palette='viridis')
plt.title("Fare amount across city")
plt.show()
```

Rendered Output:

Rendered Output:



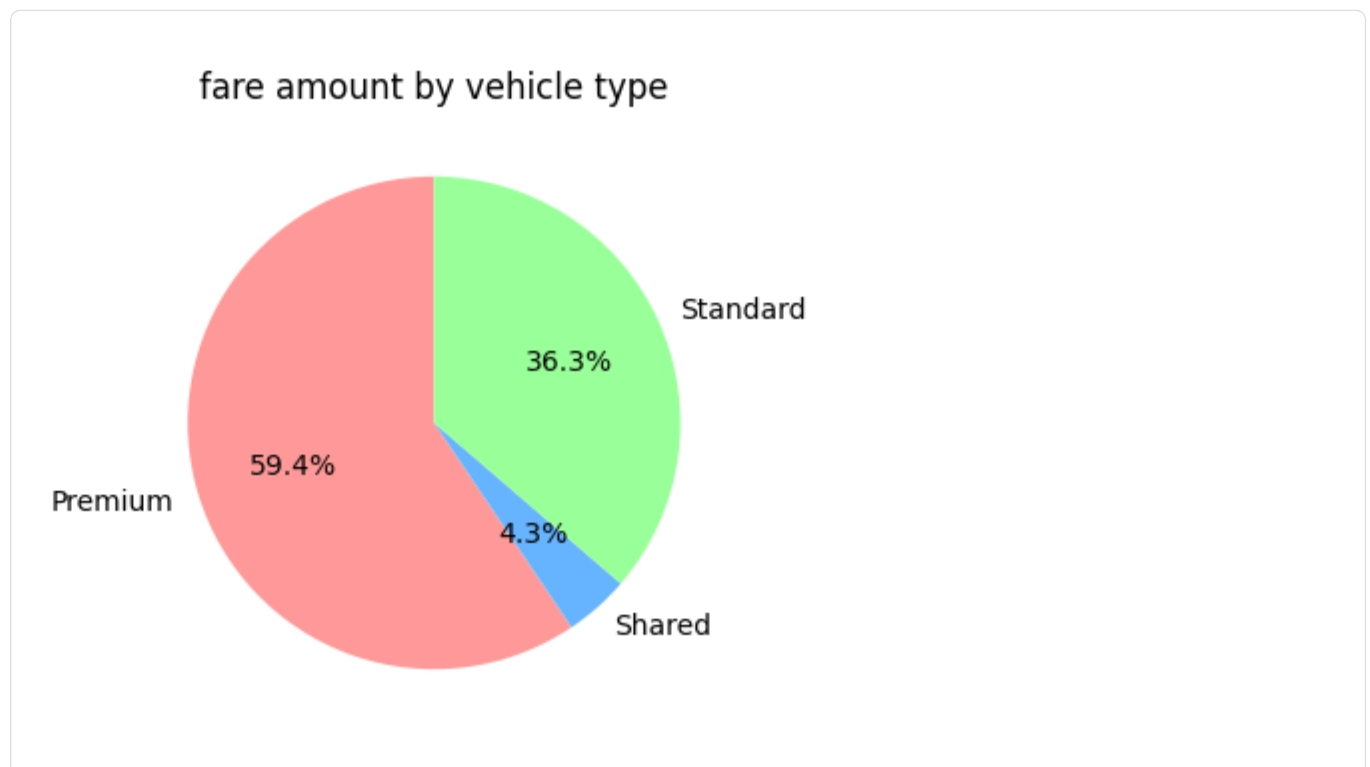
Question 3: 3. Pie Chart (Distribution)

Your Code:

```
vehicle_fares = merged_df.groupby('vehicle_type')['fare_amt'].sum().reset_index()

# Write your plotting code here:
plt.pie(vehicle_fares.fare_amt, labels=vehicle_fares.vehicle_type, autopct='%1.1f%%',
startangle=90, colors=['#ff9999','#66b3ff','#99ff99','#ffcc99'])
plt.title("fare amount by vehicle type")
plt.show()
```

Rendered Output:

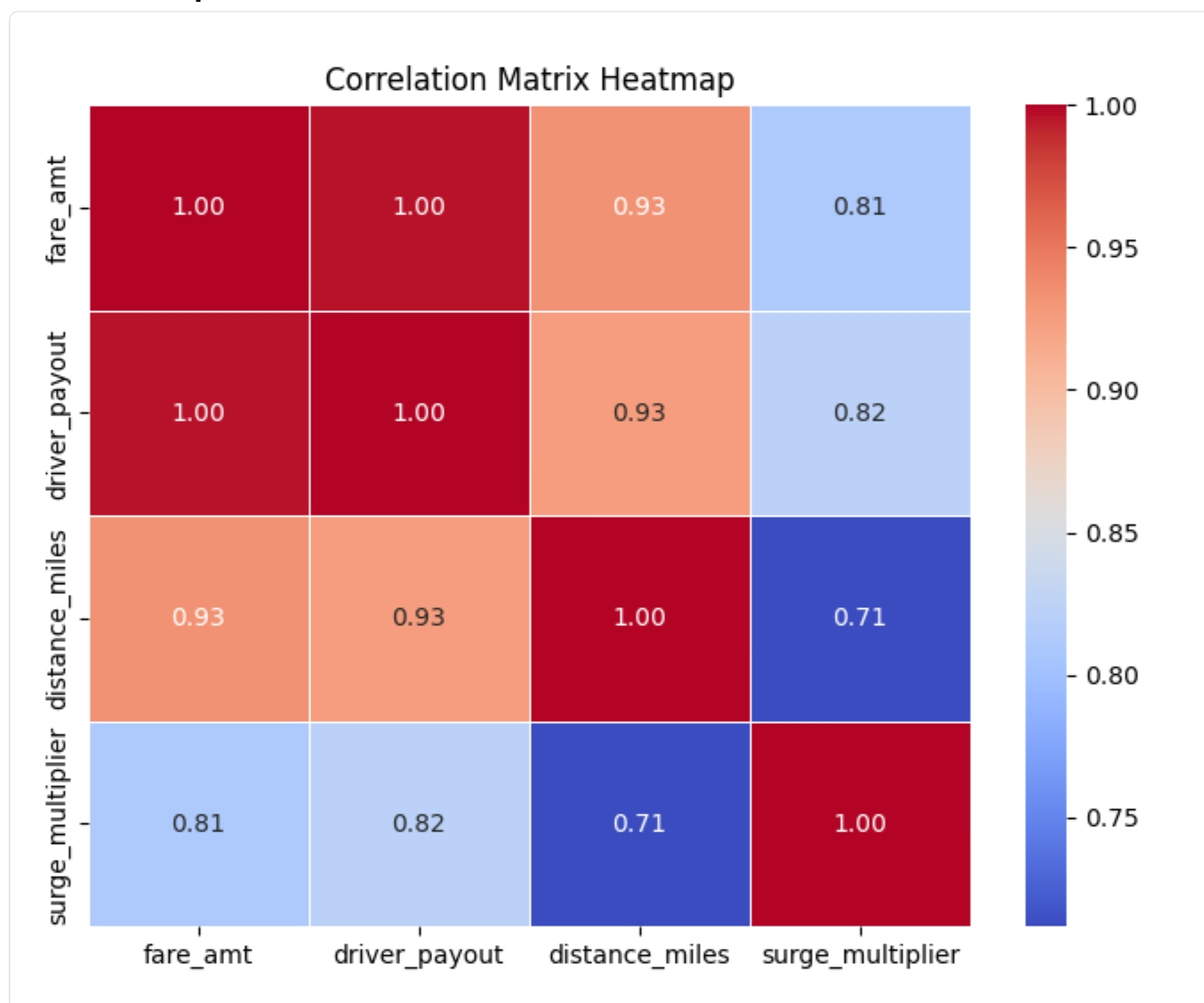


Question 4: 4. Heatmap (Correlation)

Your Code:

```
corr_cols = ['fare_amt', 'driver_payout', 'distance_miles', 'surge_multiplier']
# Write your plotting code here:
corr_matrix=merged_df[corr_cols].corr()
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f",linewidths=0.5)
plt.title("Correlation Matrix Heatmap")
plt.show()
```

Rendered Output:

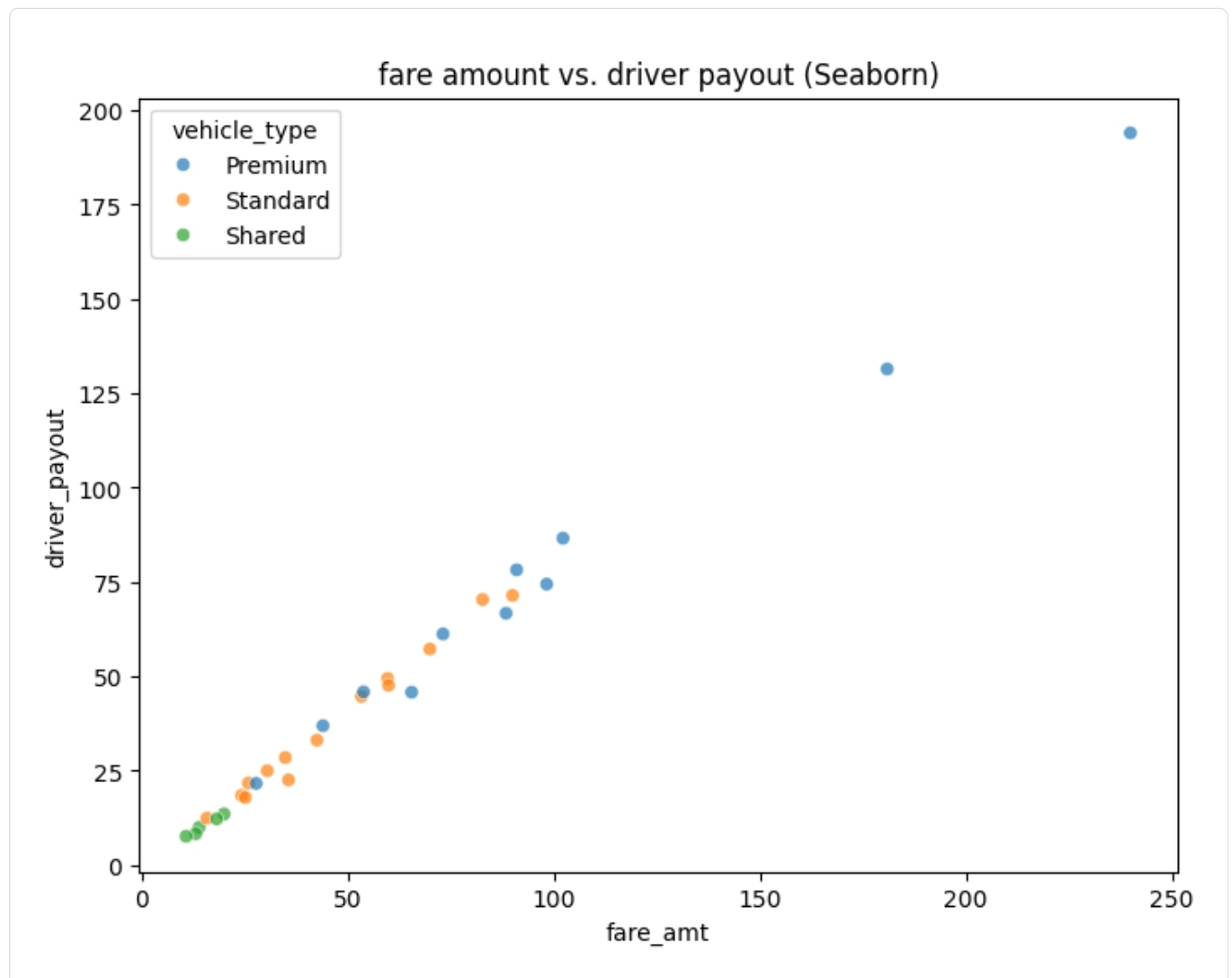


Question 5: 5. Scatter Plot (Relationships)

Your Code:

```
# Write your plotting code here:
import seaborn as sns
sns.scatterplot(data=merged_df, x='fare_amt', y='driver_payout', hue='vehicle_type', alpha=0.7)
plt.title("fare amount vs. driver payout (Seaborn)")
plt.show()
```

Rendered Output:

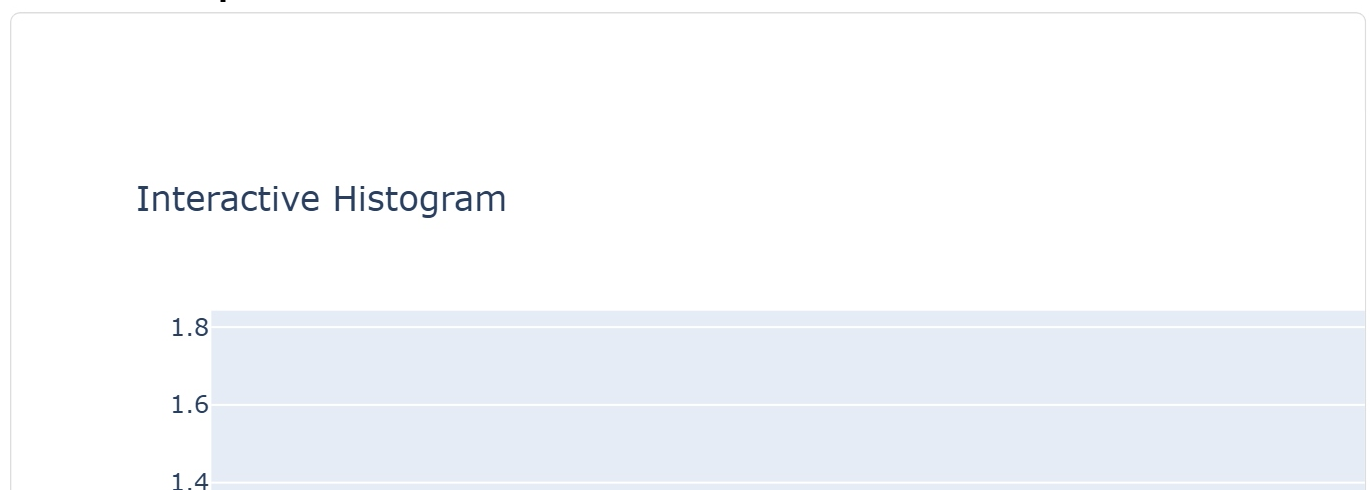


Question 6: 6. Interactive Histogram

Your Code:

```
# Write your plotting code here:  
fig = px.histogram(merged_df, x='driver_payout', nbins=30,title="Interactive Histogram")  
fig.show()
```

Rendered Output:



count

1.2

1

0.8

0.6

-1

0

1

2

3

4

driver_payout

